# Answer Key

## Other (142 Questions)

| | | | | |
|---|---|---|---|---|
| Q1. (A) | Q2. (B) | Q3. (C) | Q4. (B) | Q5. (B) |
| Q6. (D) | Q7. (B) | Q8. (C) | Q9. (D) | Q10. (A) |
| Q11. (D) | Q12. (C) | Q13. (D) | Q14. (C) | Q15. (D) |
| Q16. (C) | Q17. (B) | Q18. (C) | Q19. (B) | Q20. (D) |
| Q21. (C) | Q22. (C) | Q23. (B) | Q24. (C) | Q25. (B) |
| Q26. (C) | Q27. (B) | Q28. (B) | Q29. (B) | Q30. (C) |
| Q31. (C) | Q32. (D) | Q33. (A) | Q34. (D) | Q35. (D) |
| Q36. (B) | Q37. (B) | Q38. (D) | Q39. (D) | Q40. (B) |
| Q41. (C) | Q42. (A) | Q43. (D) | Q44. (C) | Q45. (D) |
| Q46. (D) | Q47. (B) | Q48. (B) | Q49. (C) | Q50. (D) |
| Q51. (D) | Q52. (A) | Q53. (C) | Q54. (D) | Q55. (B) |
| Q56. (A) | Q57. (B) | Q58. (B) | Q59. (C) | Q60. (C) |
| Q61. (D) | Q62. (D) | Q63. (A) | Q64. (B) | Q65. (C) |
| Q66. (B) | Q67. (A) | Q68. (D) | Q69. (A) | Q70. (B) |
| Q71. (B) | Q72. (C) | Q73. (A) | Q74. (A) | Q75. (C) |
| Q76. (D) | Q77. (A) | Q78. (B) | Q79. (A) | Q80. (C) |
| Q81. (B) | Q82. (C) | Q83. (D) | Q84. (D) | Q85. (D) |
| Q86. (C) | Q87. (C) | Q88. (D) | Q89. (D) | Q90. (C) |
| Q91. (C) | Q92. (A) | Q93. (D) | Q94. (B) | Q95. (D) |
| Q96. (B) | Q97. (A) | Q98. (B) | Q99. (D) | Q100.(A) |
| Q101.(B) | Q102.(C) | Q103.(A) | Q104.(C) | Q105.(B) |

| | | | | |
|---|---|---|---|---|
| **Q106.(A)** | **Q107.(C)** | **Q108.(B)** | **Q109.(C)** | **Q110.(D)** |
| **Q111.(B)** | **Q112.(A)** | **Q113.(B)** | **Q114.(B)** | **Q115.(C)** |
| **Q116.(A)** | **Q117.(A)** | **Q118.(B)** | **Q119.(A)** | **Q120.(C)** |
| **Q121.(B)** | **Q122.(D)** | **Q123.(C)** | **Q124.(A)** | **Q125.(B)** |
| **Q126.(C)** | **Q127.(D)** | **Q128.(A)** | **Q129.(C)** | **Q130.(D)** |
| **Q131.(A)** | **Q132.(C)** | **Q133.(D)** | **Q134.(B)** | **Q135.(D)** |
| **Q136.(C)** | **Q137.(B)** | **Q138.(A)** | **Q139.(C)** | **Q140.(D)** |
| **Q141.(B)** | **Q142.(D)** | | | |

# Solutions

**Q1. Solution**

**Correct Answer: (A)**

Energy stored in an inductor, $\mathrm{U}=\frac{1}{2} \mathrm{LI}^{2} \Rightarrow \quad \mathrm{L}=\frac{2 \mathrm{U}}{\mathrm{I}^{2}}\ [\mathrm{L}]=\frac{\left[\mathrm{ML}^{2}\ \mathrm{T}^{-2}\right]}{[\mathrm{A}]^{2}}=\left[\mathrm{ML}^{2}\ \mathrm{T}^{-2}\ \mathrm{A}^{-2}\right]$

**Q2. Solution**

**Correct Answer: (B)**

Max. K.E. $=hv-W_0$; so Max. K.E. $\propto v$

**Q3. Solution**

**Correct Answer: (C)**

Distance travelled in the nth second is given by $\mathrm{t_n}=\mathrm{u}+\frac{\mathrm{a}}{2}(2\mathrm{n}-1)$ put $\mathrm{u}=0, \mathrm{a}=\frac{4}{3}\ \mathrm{ms}^{-2}, \mathrm{n}=3$

$\therefore \quad d=0+\frac{4}{3 \times 2}(2 \times 3-1)=\frac{4}{6} \times 5=\frac{10}{3} m$

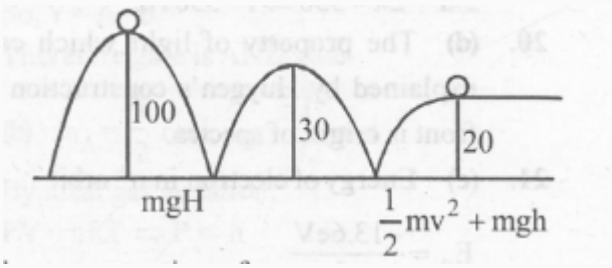
**Q4. Solution**

**Correct Answer: (B)**

Various regions of spherical shell attract the point mass inside it in various directions. These forces cancel each other completely. Therefore the gravitational force on the particle is zero.

**Q5. Solution**

**Correct Answer: (B)**




Using conservation of energy, $\mathrm{m}(10 \times 100)=\mathrm{m}\left(\frac{1}{2} \mathrm{v}^{2}+10 \times 20\right)$ or $\frac{1}{2} v^{2}=800$ or $v=\sqrt{1600}=40\ \mathrm{m/s}$

**Q6. Solution**

**Correct Answer: (D)**

Time of flight $=\frac{2 u \sin \theta}{g}=\frac{2 \times 9.8 \times \sin 30^{\circ}}{9.8}=2 \times \frac{1}{2}=1 \mathrm{sec}$

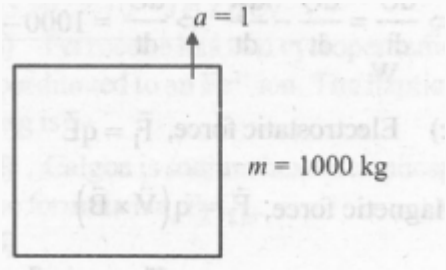
**Q7. Solution**

**Correct Answer: (B)**

Using Gauss's law, $\phi=\frac{q}{\epsilon_0}$ Here, $\mathrm{q}=$ charge inside the closed surface $\therefore \phi=\frac{q+(-2 q)+5 q}{\epsilon_0} \Rightarrow \phi=\frac{4 q}{\epsilon_0}$

**Q8. Solution**

**Correct Answer: (C)**




Total mass $= (60+940)\text{kg} = 1000$ kg Let T be the tension in the supporting cable, then
$T - 1000g = 1000 \times 1 \Rightarrow \mathrm{T} = 1000 \times 11 = 11000$ N

**Q9. Solution**

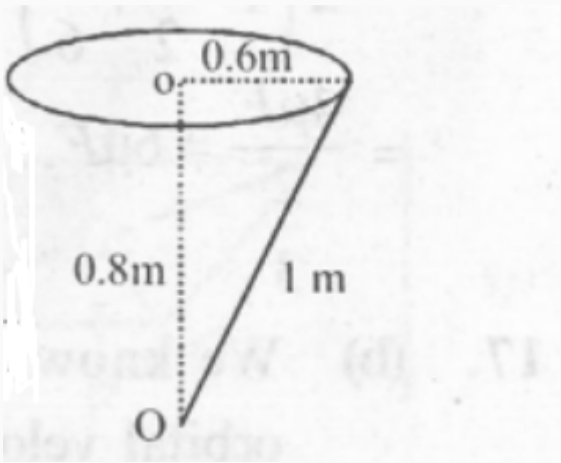
**Correct Answer: (D)**

Given : Number of cells, $n = 5$, emf of each cell $= E$ Internal resistance of each cell $= r$ In series, current through resistance $R$ $I = \frac{nE}{nr+R} = \frac{5E}{5r+R}$ In parallel, current through resistance $R$ $I' = \frac{E}{\frac{r}{n}+R} = \frac{nE}{r+nR} = \frac{5E}{r+5R}$
According to question, $I = I' \therefore \frac{5E}{5r+5R} = \frac{5E}{r+5R} \Rightarrow 5r + R = r + 5R$ or $R = r \quad \therefore \frac{R}{r} = 1$

**Q10. Solution**

**Correct Answer: (A)**

Angular momentum $L_0 = \mathrm{mvr} \sin 90^\circ = 2 \times 0.6 \times 12 \times 1 \times 1$ [ As $V = r\omega$, Sin $90^\circ = 1$] So,




$L_0 = 14.4\text{kgm}^2/\text{s}$

**Q11. Solution**

**Correct Answer: (D)**

Let $V_1$ be, the volume immersed in water. $V_2$ be the volume immersed in oil. In equilibrium condition,
$v_1\rho_w g + v_2\rho_0 g = (v_1+v_2)\rho_e g$ $v_1 + \frac{v_2\rho_0}{\rho_w} = (v_1+v_2)\frac{\rho_c}{\rho_w} \Rightarrow v_1 + 0.8v_2 = 0.9v_1 + 0.9v_2 \Rightarrow 0.1v_1 = 0.1v_2$
$\Rightarrow v_1 : v_2 = 1 : 1$ ~

**Q12. Solution**

**Correct Answer: (C)**

Electrostatic force, $\overrightarrow{\mathrm{F}}_1 = \mathrm{q}\overrightarrow{\mathrm{E}}$ Magnetic force, $\overrightarrow{\mathrm{F}}_2 = \mathrm{q}(\overrightarrow{\mathrm{V}} \times \overrightarrow{\mathrm{B}})$ ~

**Q13. Solution**

**Correct Answer: (D)**

Given, Rate of heat supplied, $\frac{\mathrm{dQ}}{\mathrm{dt}} = 1000 \mathrm{~W}$ Rate of work performed, $\frac{\mathrm{dw}}{\mathrm{dt}} = 200 \mathrm{~W}$ Using first law of thermodynamics $dQ = dU + dw \Rightarrow \frac{\mathrm{dU}}{\mathrm{dt}} = \frac{\mathrm{dQ}}{\mathrm{dt}} - \frac{\mathrm{dw}}{\mathrm{dt}} \Rightarrow \frac{\mathrm{dU}}{\mathrm{dt}} = 1000 - 200 = 800$ ,

**Q14. Solution**

**Correct Answer: (C)**

Averageemf, $\mathrm{e} = \frac{\text{Changein flux}}{\text{Time}} = -\frac{\Delta\phi}{\Delta \mathrm{t}} = -\frac{0-(4\times(2.5\times2)\cos 60^\circ)}{10} = +1 \mathrm{~V}$ ,

**Q15. Solution**

**Correct Answer: (D)**

de-Broglie wavelength, $\lambda = \frac{\mathrm{h}}{p} = \frac{h}{\sqrt{2\cdot m\cdot(K\cdot E)}} \quad \therefore \lambda \propto \frac{1}{\sqrt{K\cdot E}}$ If K.E is doubled, $\lambda$ becomes $\frac{\lambda}{\sqrt{2}}$ .

**Q16. Solution**

**Correct Answer: (C)**

We have, $C_i = \frac{A\epsilon_0}{d} = 4\mu F \; C_f = \frac{A\epsilon_0}{d-t+\frac{t}{k}} = \frac{A\epsilon_0}{d-\frac{d}{2}+\frac{d}{2\times3}} = \frac{A\in_0}{d\left(1-\frac{1}{2}+\frac{1}{6}\right)} = \frac{4\mu F}{\frac{2}{3}} = 6\mu F$ ^

**Q17. Solution**

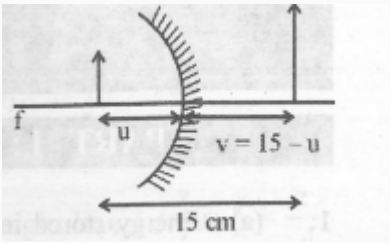
**Correct Answer: (B)**

We know that $v_e = \sqrt{2}v_0$, where $\mathrm{v}_0$ is orbital velocity. K.E. in the orbit, $E = \frac{1}{2}M{v_0}^2$ K.E. to escape $E = \frac{1}{2}Mv_e^2 = \frac{1}{2}M\left(2{v_0}^2\right) = \frac{1}{2}Mv_0^2 \times 2 = 2E$ ^

**Q18. Solution**

**Correct Answer: (C)**

Since, image is magnified. So, it is a concave mirror. $\therefore \mathrm{m}=2=\frac{-\mathrm{v}}{\mathrm{u}}$ $2=\frac{-(15-u)}{-u}$ $2u=15-u$
$3\mathrm{u}=15 \Rightarrow \mathrm{u}=5$ cm $\mathrm{v}=15-\mathrm{u}=15-5=10$ cm Bymirror formula, $\frac{1}{\mathrm{f}}=\frac{1}{\mathrm{v}}+\frac{1}{\mathrm{u}}$




$=\frac{1}{10}+\frac{1}{(-5)}=\frac{1-2}{10}=\frac{-1}{10}$ $\mathrm{f}=-10$ cm

^

**Q19. Solution**

**Correct Answer: (B)**

Equation of wave is, $y=2\cos 2\pi(330t-x)$ m $y=A\cos(\omega t-kx)$ On comparing $\omega=2\pi\times 330$
$2\pi\mathrm{f}=2\pi\times 330 \Rightarrow \mathrm{f}=330$ Hz $[\because \omega=2\pi$ F$]$ !

**Q20. Solution**

**Correct Answer: (D)**

The property of light which cannot be explained by Huygen's construction of wave front is origin of spectra. ,

**Q21. Solution**

**Correct Answer: (C)**

Young's modulus depends on the material not depends on length and cross sectional area. So Young's modulus remains constant.

**Q22. Solution**

**Correct Answer: (C)**

Energy of electron in $\mathrm{n}^{\mathrm{th}}$ orbit $E_n=\frac{-13.6eV}{n^2}$ Excitation energy of electron from $n=1$ to $n=2$
$E=-13.6\left[\frac{1}{2^2}-\frac{1}{1}\right]=-13.6\left[\frac{1}{4}-\frac{1}{1}\right]=10.2\mathrm{eV}$

**Q23. Solution**

**Correct Answer: (B)**

According to Brewester's law if the reflected ray is completely polarized, then the reflected and refracted ray is perpendicular to each other. Thus, $\theta=90^{\circ}$

**Q24. Solution**

**Correct Answer: (C)**

In the breakdown region, a reverse biased zener diode works as a voltage regulator.

**Q25. Solution**

**Correct Answer: (B)**

The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in range $30 < \mathrm{A} < 17^{\circ}$. So, $26^{\mathrm{Fe}^{56}}$ has maximum binding energy per nucleon.

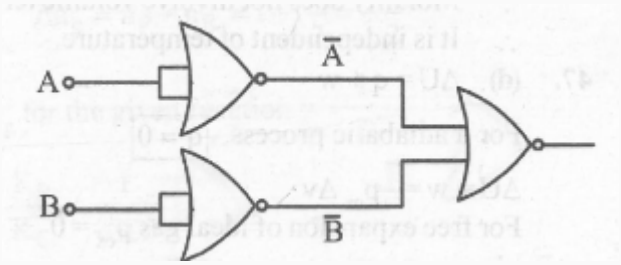
**Q26. Solution**

**Correct Answer: (C)**

Since $\frac{F-32}{9} = \frac{C}{5} \Rightarrow \Delta \mathrm{C} = \frac{5}{9} \Delta \mathrm{F} \Rightarrow 40 = \frac{5}{9} \Delta \mathrm{F} \Rightarrow \Delta \mathrm{F} = 72^{\circ} \mathrm{F}$

**Q27. Solution**

**Correct Answer: (B)**




$\mathrm{Y} = \overline{\overline{\mathrm{A}} + \overline{\mathrm{B}}} = \overline{\overline{\mathrm{A}}} \cdot \overline{\overline{\mathrm{B}}} = \mathrm{A}.\mathrm{B}$ So, $\mathrm{Y} = \mathrm{A} \cdot \mathrm{B}$ Therefore gate is AND gate.

**Q28. Solution**

**Correct Answer: (B)**

$\mathrm{n_A} = \frac{1}{2}$ mol, $\mathrm{n_B} = \frac{1}{32}$ mol By ideal gas equation, $PV = nRT \Rightarrow P \propto n \therefore \frac{\mathrm{P_A}}{\mathrm{P_B}} = \frac{\mathrm{n_A}}{\mathrm{n_B}} = \frac{32}{2} = 16$

**Q29. Solution**

**Correct Answer: (B)**

Rising half to peak $t = T/6$ t $= \frac{2\pi}{6\omega} = \frac{\pi}{3\omega} = \frac{\pi}{300\pi} = \frac{1}{300} = 3.3$ ms Here $\omega = 100\pi \mathrm{rad/s}$

**Q30. Solution**

**Correct Answer: (C)**

$\lambda = 3000\tilde{A} \ldots = 3 \times 10^{-7}$ m, $\phi = 2.13$eV K.E. $= hv - hv_0 = hv - \phi = \frac{hc}{\lambda} - \phi$
$= \frac{(6.626 \times 10^{-34} \times 3 \times 10^{8})}{3 \times 10^{-7}} - (2.13 \times 1.6 \times 10^{-19}) = \frac{1.98 \times 10^{-25}}{3 \times 10^{-7}} - (3.408 \times 10^{-19})$
$= (6.60 \times 10^{-19}) - (3.408 \times 10^{-19}) = 3.192 \times 10^{-19}$ J $= 2.0$eV

**Q31. Solution**

**Correct Answer: (C)**

Time period of simple pendulum, $\mathrm{T}=2\pi\sqrt{\frac{l}{\mathrm{g}}}$ & $\mathrm{g}=\frac{\mathrm{GM}}{(\mathrm{R}+\mathrm{h})^2}$ $\therefore \mathrm{T} \propto (\mathrm{R}+\mathrm{h})$

$\Rightarrow \frac{\mathrm{T}_1}{\mathrm{T}_2}=\frac{(\mathrm{R}+\mathrm{h}_1)}{(\mathrm{R}+\mathrm{h}_2)}=\frac{(\mathrm{R}+\mathrm{R})}{(\mathrm{R}+2\mathrm{R})}=\frac{2}{3} \Rightarrow 3\ \mathrm{T}_1=2\ \mathrm{T}_2$

**Q32. Solution**

**Correct Answer: (D)**

Calgon is sodium hexametaphosphate with the formula $\mathrm{Na}_6\mathrm{P}_6\mathrm{O}_{18}$ :

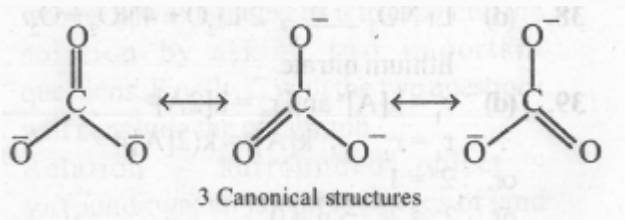
**Q33. Solution**

**Correct Answer: (A)**

Ferrocene has two cyclopentadienyl rings coordinated to an $\mathrm{Fe}^{2+}$ ion. The hapticity of each ring is 5 .

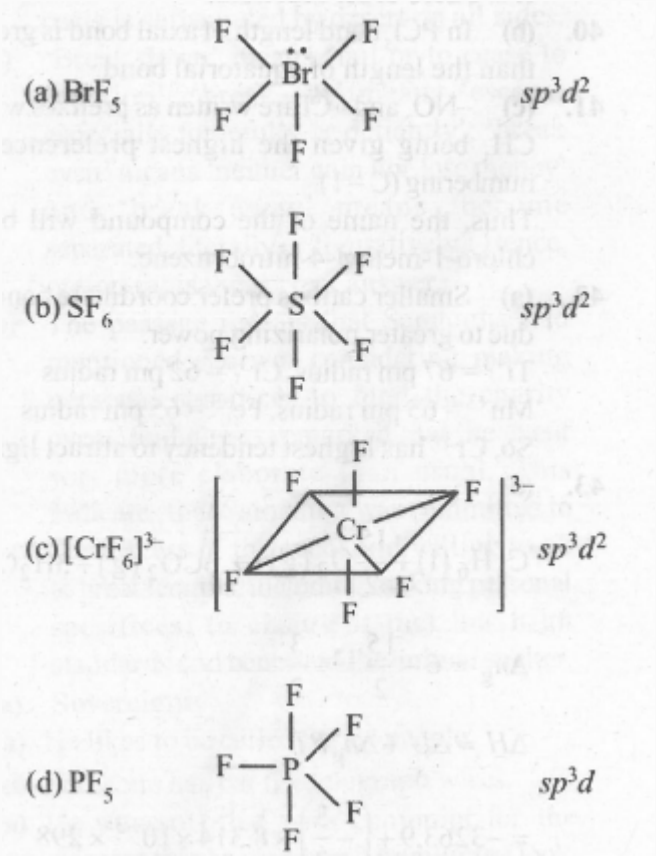
**Q34. Solution**

**Correct Answer: (D)**



**Q35. Solution**

**Correct Answer: (D)**




**Q36. Solution**

**Correct Answer: (B)**

As electron of charge ' $e$ ' is passed through ' $V$ ' volt, kinetic energy of electron will be eV . Wavelength of electron wave $(\lambda)=\frac{h}{\sqrt{2m \cdot \mathrm{KE}}}$ $\lambda=-\frac{h}{\sqrt{2 m e \mathrm{V}}}$ $\quad \therefore \frac{h}{\lambda}=\sqrt{2 m e \mathrm{V}}$

**Q37. Solution**

**Correct Answer: (B)**

- Maltose is composed of two glucose units. - Sucrose has glycosidic linkage between $\mathrm{C}_1$ of $\alpha$-glucose and $C_2$ of $\beta$-fructose. - Lactose is composed of galactose and glucose - Amylose has glycosidic linkage between $C_1$ of $\alpha$-glucose $C_4$ of $\alpha$-glucose.

**Q38. Solution**

**Correct Answer: (D)**

$\mathrm{LiNO}_3 \xrightarrow{\Delta} 2\mathrm{Li}_2\mathrm{O}+4\mathrm{NO}_2+\mathrm{O}_2$ lithium nitrate

**Q39. Solution**
**Correct Answer: (D)**

$r_1 = k[A]^n$ and $r_2 = k[2A]^n \because \quad \mathrm{r}_1 = \mathrm{r}_2 \quad \therefore \mathrm{k}[\mathrm{A}]^{\mathrm{n}} = \mathrm{k}(2[\mathrm{A}])^{\mathrm{n}}$ or, $\quad 2^n = 1$ or, $2^{\mathrm{n}} = 2^{\circ} \Rightarrow \mathrm{n} = 0$ It is a zero order reaction.

**Q40. Solution**
**Correct Answer: (B)**

In $\mathrm{PCl}_5$ bond length of axial bond is greater than the length of equatorial bond.

**Q41. Solution**
**Correct Answer: (C)**

$-\mathrm{NO}_2$ and -Cl are written as prefixes with $\mathrm{CH}_3$ being given the highest preference for numbering $(\mathrm{C}-1)$. Thus, the name of the compound will be $2-$ chloro-1-methyl-4-nitrobenzene.

**Q42. Solution**
**Correct Answer: (A)**

smaller cations prefer coordinate bonding due to greater polarizing power. $\mathrm{Ti}^{+3} = 67\mathrm{pm}$ radius, $\mathrm{Cr}^{3+} = 62\mathrm{pm}$ radius $\mathrm{Mn}^{+3} = 65\mathrm{pm}$ radius, $\mathrm{Fe}^{+3} = 65\mathrm{pm}$ radius So, $\mathrm{Cr}^{3+}$ has highest tendency to attract ligand.

**Q43. Solution**
**Correct Answer: (D)**

$\mathrm{C}_6\mathrm{H}_6(\mathrm{I}) + \frac{15}{2}\mathrm{O}_2(\mathrm{~g}) \rightarrow 6\mathrm{CO}_2(\mathrm{~g}) + 3\mathrm{H}_2\mathrm{O}(\mathrm{l})\ \Delta n_g = 6 - \frac{15}{2} = -\frac{3}{2}\ \Delta H = \Delta U + \Delta n_{\mathrm{g}} RT$
$= -3263.9 + \left(-\frac{3}{2}\right) \times 8.314 \times 10^{-3} \times 298 = -3263.9 + (-3.71) = -3267.6 \text{ kJ mol}^{-1}$

**Q44. Solution**
**Correct Answer: (C)**

As the concentration of reactant decreases from 0.8 to 0.4 in 15 minutes, hence the $t_{1/2}$ is 15 minutes. To fall the concentration from 0.1 to 0.025 , we need two half lives, i.e., 30 minutes.

**Q45. Solution**
**Correct Answer: (D)**

Molality-Moles of solute/mass of solvent in kg $\because \quad$ Molality does not involve volume term. $\therefore \quad$ It is independent of temperature.

**Q46. Solution**

**Correct Answer: (D)**

$\Delta U = q + w$ For a adiabatic process, $\mathrm{q} = 0$ $\Delta \mathrm{U} = \mathrm{w} = -\mathrm{p}_{\mathrm{ex}} \Delta \mathrm{v}$ For free expansion of ideal gas $\mathrm{p}_0 = 0$ $\Delta \mathrm{U} = \mathrm{w} = 0$ As $\Delta \mathrm{U} = \mathrm{C} \Delta \mathrm{T} = 0$ $\Delta \mathrm{T} = 0$

**Q47. Solution**

**Correct Answer: (B)**

The reaction of alcohol with Lucas reagent is mostly an $\mathrm{S}_N 1$ mechanism reaction and the rate of reaction is directly proportional to the carbocation stability formed in the reaction, hence tertiary alcohol will react fastest.

**Q48. Solution**

**Correct Answer: (B)**

In compounds (I), (III) and (IV) the -M effect and hyperconjugation effect are inoperative at meta positions. So we should consider the inductive effect of the groups. -I effect will decrease the stability of the free radical by withdrawing the electrons toward it itself. The order of -I effect is $\mathrm{NO}_2 > \mathrm{CN} > \mathrm{OH}$. Therefore (III) is most destabilised. In compound (II) the OH act as electron donating group through +M effect which is operative at para position. This increase the electron density at para position and stabilise the free radical. Hence compound (1) is most stabilised. The order of stability of free radical is $\mathrm{II} > \mathrm{I} > \mathrm{IV} > \mathrm{III}$

**Q49. Solution**

**Correct Answer: (C)**

The complex is a cationic complex with two ammine $(\mathrm{NH}_3)$, one chloro $(\mathrm{Cl})$ and one methanamine ligand. Thus, the name will be :Diamminechloro (methanamine) platinum (II) chloride.

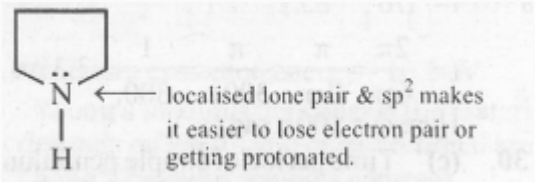
**Q50. Solution**

**Correct Answer: (D)**

Colligative property depends on the number of particles or ions. Option (d) consists 4 of ions due to which it shows highest colligative property.

**Q51. Solution**

**Correct Answer: (D)**



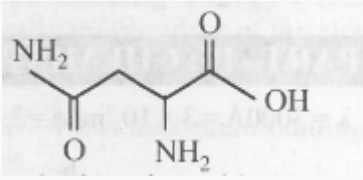
**Q52. Solution**

**Correct Answer: (A)**

$\text{Calamine} \rightarrow ZnCO_3$
$\text{Malachite} \rightarrow CuCO_3Cu(OH)_2$
$\text{Magnetite} \rightarrow Fe_3O_4$
$\text{Cryolite} \rightarrow Na_3AlF_6$

**Q53. Solution**

**Correct Answer: (C)**




Asparagine has only one basic functional group in its chemical structure. Others are basic amino acid with more than one basic functional group.

**Q54. Solution**

**Correct Answer: (D)**

$CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$ $K_P$ and $K_C$ are related as $K_P = K_C(RT)^{\Delta n_g}$
$\Rightarrow \quad \Delta n_g = n_P - n_R = 1 - 1\frac{1}{2} = -\frac{1}{2}$ $\Delta n_g$ for the given reaction $= -\frac{1}{2} \therefore \frac{K_P}{K_C} = \frac{1}{\sqrt{RT}}$

**Q55. Solution**

**Correct Answer: (B)**

Mn shows highest oxidation state of +7 in 3d series metals. Ti; Co and Fe shows highest. Oxidation state of $+4, +4$ and +6 respectively. Oxidation state of $+4, +4$ and +6 respectively.

**Q56. Solution**

**Correct Answer: (A)**

Histamine is a potent vasodilator. It contracts smooth muscles and relaxes other muscles of blood vessels of stomach walls. In stomach, histamine stimulates the secretion of hydrochloric acid and pepsin.

**Q57. Solution**

**Correct Answer: (B)**

$As_2O_3 + 3H_2S \rightarrow As_2S_3 + 3H_2O$

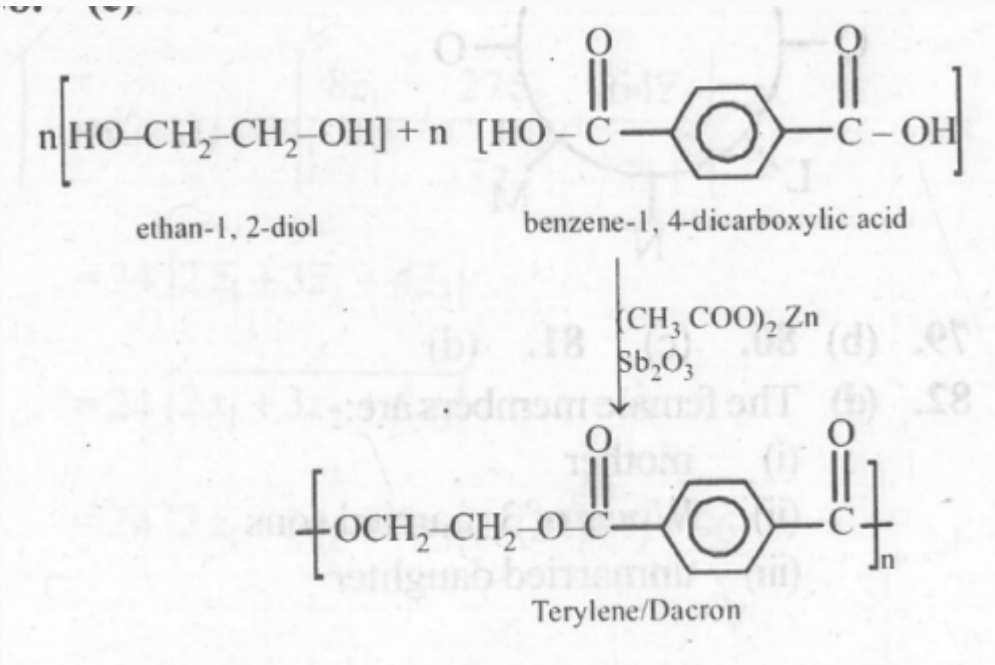
**Q58. Solution**

**Correct Answer: (B)**

Complex with maximum number of unpaired electron will exhibit maximum attraction to an applied magnetic field. $[Zn(H_2O)_6]^{2+} \rightarrow d^{10}$ system $\rightarrow t_{2g}^6 e_g^4$ 0 unpaired $e^-$ $[Co(H_2O)_6]^{2+} \rightarrow d^7$ system $\rightarrow t_{2g}^5 e_g^2$ 3 unpaired $e^-$ $[Co(en)_3]^{3+} \rightarrow d^6$ system $\rightarrow t_{2g}^6 e_g^0$ 0 unpaired $e^-$ $[Ni(H_2O)_6]^{2+} \rightarrow d^8$ system $\rightarrow t_{2g}^6 e_g^2$ 2 unpaired e

**Q59. Solution**

**Correct Answer: (C)**




**Q60. Solution**

**Correct Answer: (C)**

$N_2(g) + O_2(g) \rightarrow 2NO(g)$ $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ Higher concentration of $NO_2$ damage the leaves and retard the rate of photosynthesis.

**Q61. Solution**

**Correct Answer: (D)**

All the sentences except D are following some other sentences as they appear to be linked with something that has been said earlier. So, D will be the introductory sentence. B will follow D as it correctly makes a link with the 'priority' mentioned in D. A will come next as it explains why 'this planning (mentioned in B)' may go wrong. E will fall next as it tries to find the solution by asking two important questions. Finally, C with the two questions will conclude the paragraph.

**Q62. Solution**
**Correct Answer: (D)**

Relation - surrounded object surroundings; an island is a piece of land surrounded by water on all sides as an oasis is surrounded by desert on all sides.

**Q63. Solution**
**Correct Answer: (A)**

'Break down' means 'fail or to cease to function'; 'break out' means 'escape, especially forcefully or defiantly'; 'break even' means 'neither gain nor lose money' and 'break away' means 'become separated, literally or figuratively'. Hence, option (1) is the correct answer.

**Q64. Solution**
**Correct Answer: (B)**

The passage reveals that Samantha had mentioned she was considering making personal sacrifices to fund the charity event, and guests remarked that the event was more elaborate than usual. This indicates that Samantha was committed to the success of the event and willing to go to great lengths, including making personal sacrifices, to ensure it met her high standards and benefited the animal shelter.

**Q65. Solution**
**Correct Answer: (C)**

The error is in part (3). The phrase "is used to speak" is incorrect. The correct structure should be "is used to speaking" because "used to" is followed by the gerund form (verb + -ing) when indicating a habitual action. So the corrected sentence should be: My friend is used to speaking Malayalam with his friends. Thus, (3) contains the error.

**Q66. Solution**
**Correct Answer: (B)**

The correct word is "derisive", which means showing contempt or ridicule. In this context, it fits because "contempt" indicates a sense of disdain or mockery, which aligns with a "derisive" look. "Humiliated" (Option 1) is incorrect because it describes the feeling of the prisoner, not the look given to him. "Pitying" (Option 3) is incorrect because pity suggests sympathy, which contrasts with contempt. "Hurtful" (Option 4) is not the best fit, as it focuses on causing pain, whereas "derisive" is specifically about mockery or scorn. Thus, "derisive" is the best choice.

**Q67. Solution**
**Correct Answer: (A)**

To convert the sentence from Active to Passive voice: Active voice: He likes people to call him Sir. Passive voice: He likes to be called Sir by people. Explanation: The structure of the passive sentence follows: [Subject] + [likes] + [to be + past participle] + [by + agent]. In this case, "He" is the subject, "likes" remains unchanged, and the verb "to call" is changed to the passive form "to be called." Thus, (1) is the correct answer.

**Q68. Solution**

**Correct Answer: (D)**

Solution: The given sentence is in Passive Voice, and we need to convert it into Active Voice. Passive voice: The telegraph wires have been cut. Active voice: Someone has cut the telegraph wires. In Passive voice, the subject (telegraph wires) receives the action, while in Active voice, the subject (someone) performs the action. Explanation: Option (1) is incorrect because it has the wrong structure: "has been cut" should be in the active voice. Option (2) is incorrect because it implies that no one has cut the wires, which contradicts the original meaning. Option (3) is incorrect because it implies that the telegraph wires did the cutting, which is nonsensical. Thus, (4) is the correct answer.

**Q69. Solution**

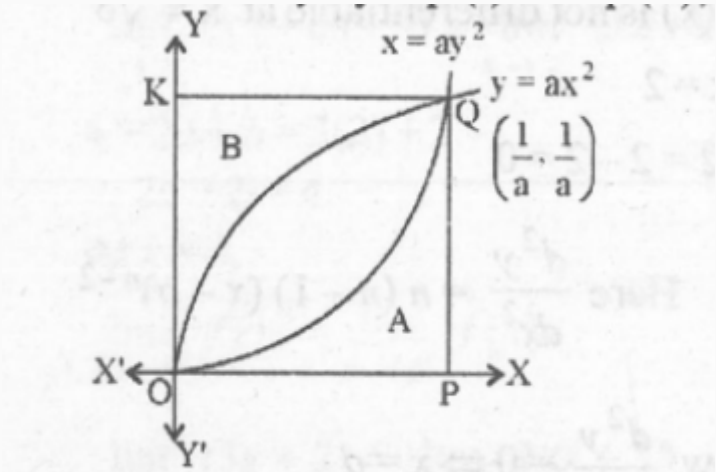
**Correct Answer: (A)**

The correct spelling is "Sovereignty", which refers to the authority of a state to govern itself or another state. Incorrect spellings: Soveriegnty ❌ (Incorrect order of letters) Sovereignity ❌ (Incorrect suffix) Soveriegnity ❌ (Incorrect order of letters) Thus, "Sovereignty" is the correct answer.

**Q70. Solution**

**Correct Answer: (B)**

We have given, $y = ax^2$ ....(i) and $x = ay^2$ .....(ii) Put the value of $y$ by Eq. (i) in Eq. (ii), we get




$x = a \times a^2x^4 \Rightarrow x^4a^3 - x = 0\ x\left(x^3a^3 - 1\right) = 0 \Rightarrow x = 0, \frac{1}{2}$ When, $x = 0 \Rightarrow y = 0$ and $x = \frac{1}{a} \Rightarrow y = \frac{1}{a}$ Here, points of intersection of curves $y = ax^2$ and $x = ay^2$ are $(0, 0)$ and $\left(\frac{1}{a}, \frac{1}{a}\right)$. $\therefore$ Required area

$A = \int_{x=a}^{x=b} [f_2(x) - f_1(x)]dx\ 3 = \int_0^{1/a} \left(\frac{\sqrt{x}}{\sqrt{a}} - ax^2\right)dx\ 3 = \left[\frac{1}{\sqrt{a}} \times \frac{2}{3}x^{3/2} - \frac{ax^3}{3}\right]_0^{1/a}$

$3 = \frac{2}{3\sqrt{a}}\left[\left(\frac{1}{a}\right)^{3/2}\right] - \frac{a}{3}\left[\left(\frac{1}{a}\right)^3\right]\ 3 = \frac{2}{3\sqrt{a}} \times \frac{1}{a\sqrt{a}} - \frac{a}{3} \times \frac{1}{a^3}\ 3 = \frac{2}{3a^2} - \frac{1}{3a^2} \Rightarrow 3 = \frac{2-1}{3a^2} \Rightarrow 9a^2 = 1$

$a^2 = \frac{1}{9} \Rightarrow a = \frac{1}{3}$

**Q71. Solution**

**Correct Answer: (B)**

The passage discusses how urban gardening allows city residents to grow their own produce, thereby providing fresh and organic food, fostering community relationships, and enhancing environmental sustainability. It highlights the benefits of urban gardening and acknowledges the challenges faced but does not emphasize it as merely decorative, expensive, or a practice for rural individuals seeking city experiences. Therefore, option $B$ best encapsulates the main points of the passage.

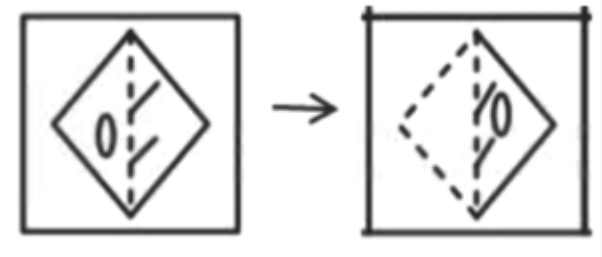
**Q72. Solution**

**Correct Answer: (C)**

The problem requires identifying the correct option figure that completes the sequence of question figures. Step-by-step approach: Look for a pattern in shapes: The sequence alternates between triangles, arrows, and other shapes such as stars and line segments. Notice how the first two symbols in each segment seem to repeat similarly after two positions. Observe movement and transformation: The triangles are either pointing left, right, or upwards. Their positions and orientations change from one figure to the next. The asterisk symbol (*) appears in alternating positions between the first and second positions. Check for repeated patterns: The arrangement seems to shift symbols in a predictable manner, with some rotation of the triangle elements. Options analysis: By comparing each option ( $1, 2, 3, 4$ ) to the sequence, look for the one that continues the transformations logically. Solution: The correct answer is Option (3) 3. The reasoning: Option 3 follows the established sequence of alternating and rotating symbols with correct placement of stars and triangles in the correct orientations.

**Q73. Solution**

**Correct Answer: (A)**

After folding, the sheet will look like the below figure: 

**Q74. Solution**

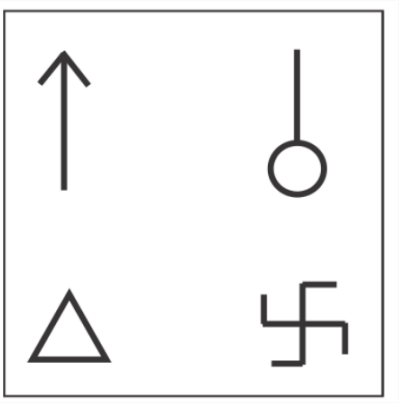
**Correct Answer: (A)**

In the given diagram, we can see that in the second figure, the diagonal elements i.e top left and bottom right gets inverted compared to the first figure and other diagonal elements are fixed.

Again, in the third figure the diagonal elements i.e bottom left and top right gets inverted and others are fixed.

Therefore, the diagonal elements gets inverted alternately.

Therefore, the answer is

`  `

`.

**Q75. Solution**

**Correct Answer: (C)**

These groups of numbers are in pattern of dates(mm - dd - yy). The days on these dates are: 5 - Jan - 96 to 27 - Jan - 96 $\rightarrow$ 23 days 27 - Jan - 96 to 18 - Feb - 96 $\rightarrow$ 23 days 18 - Feb - 96 to 11 - Mar - 96 (not 12 - Mar - 96) $\rightarrow$ 23 days 11 - Mar - 96 (not 12 - Mar - 96) to 2 - April - 96 $\rightarrow$ 23 days Hence, '12 - 03 $-$ 96 ' does not fit the series.

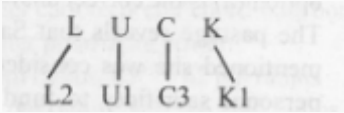
**Q76. Solution**

**Correct Answer: (D)**

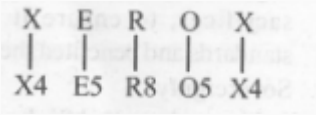
18 K 3 Q7 J2T8 18 $\div 3+7 \times 2-8$ $6+14-8$ $20-8=12$

**Q77. Solution**

**Correct Answer: (A)**

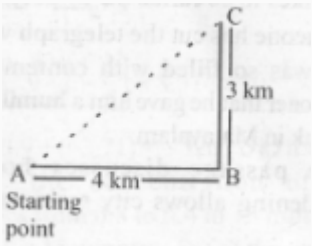
As 

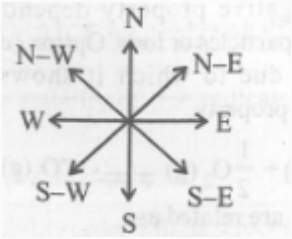

where the place value of L is $12$, U is $21$, C is 3 and K is 11 , So the last number is written with the present letter.

Same as, 


**Q78. Solution**

**Correct Answer: (B)**




according to the question. by Pythagoras theorem, $\mathrm{AC}^2 = \mathrm{BC}^2 + \mathrm{AB}^2$ $\mathrm{AC}^2 = 3^2 + 4^2$ $AC^2 = 9 + 16$ $\mathrm{AC} = \sqrt{25} = 5\mathrm{Km}$ So, Raju is 5 Km far from his initial point and Raju is North-East direction from initial point.

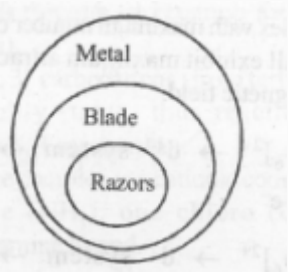
**Q79. Solution**

**Correct Answer: (A)**

All blue are shades. Some shades are colors. Since all colors are shades and all blue are colors, it follows that all blue are indeed shades (Conclusion 1). Also, since all colors are shades, it implies that some shades are indeed colors (Conclusion 2).

**Q80. Solution**

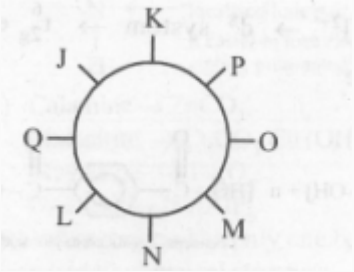
**Correct Answer: (C)**




only conclusion 2 follows.

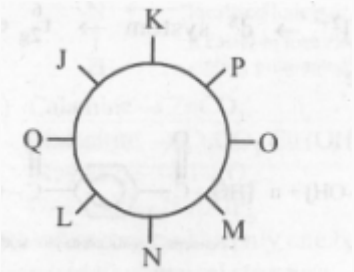
**Q81. Solution**

**Correct Answer: (B)**

Seating arrangement is as follows: 

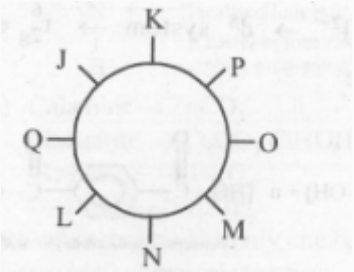

**Q82. Solution**

**Correct Answer: (C)**

Seating arrangement is as follows: 


**Q83. Solution**

**Correct Answer: (D)**

Seating arrangement is as follows: 

**Q84. Solution**
**Correct Answer: (D)**

The female members are:- (i) mother (ii) Wives of 3 married sons (iii) unmarried daughter (iv) 2 daughter of each of two sons $\therefore$ Total No of females $= 1 + 3 + 1 + 2 \times 2 = 9$

**Q85. Solution**
**Correct Answer: (D)**

The colour of human blood is red. Here white means red. Therefore white is our answer. Do not opt for black because red means black implies that black is called red.

**Q86. Solution**
**Correct Answer: (C)**

Sister of my brother $=$ My sister Father of my sister = My father Daughter of my father $=$ My sister

**Q87. Solution**
**Correct Answer: (C)**

Given: $\mathrm{M_NO_MNNN_P_NO_PMM_OPP_}$ By checking options and substituting accordingly. 1) $\mathrm{MPMOONP} \rightarrow \mathrm{M\underline{M}NO\underline{P}/MNN\underline{M}/\underline{O}NO\underline{O}P/MN\underline{OP\underline{P}}}$ The pattern followed here is:- M M NOP/MNNOP/ MNOOP/MNOPP Here, the letters $M, N, O,$ and $P$ are repeating respectively ($MM, NN, OO, PP$) which makes the series complete. 4) $\mathrm{MPMONOP} \rightarrow \mathrm{M\underline{M}NO\underline{P}/M}N\ \mathrm{N\underline{M}P/\underline{O}NO\underline{N}P/M\underline{O}}OP\underline{\mathrm{P}}$ Hence, 'MPOMONP' is the correct answer.

**Q88. Solution**
**Correct Answer: (D)**

According to the questions $\mathrm{F} > \mathrm{D} > \mathrm{B} > \mathrm{A}$ or C Therefore either A or C scored the least So the answer is (4)

**Q89. Solution**
**Correct Answer: (D)**

$\mathrm{B} > \mathrm{C} > \mathrm{D} > \mathrm{E} > \mathrm{A}$ $\quad 92\% \quad 71\%$ B scored more than D. There-fore, B must have scored more than $97\%$ marks.

**Q90. Solution**
**Correct Answer: (C)**

In fig $(A), (3 \times 3) + (6 \times 5) = 39$ In fig $(B), (4 \times 4) + (5 \times 7) = 51$ $\therefore$ In fig $(\mathrm{C})$, missing number $= (3 \times 4) + (5 \times 5) = 37$

**Q91. Solution**

**Correct Answer: (C)**

Area common to $\bigcirc$ and $\Delta$.

**Q92. Solution**

**Correct Answer: (A)**

- TOMATO $\rightarrow$ MTOOTA - The transformation involves rearranging the letters of "TOMATO": - The first letter T moves to the second position. - The second letter O moves to the third position. - The third letter M stays in the same position. - The fourth letter A moves to the fifth position. - The fifth letter T moves to the sixth position. - The sixth letter O moves to the first position. Now, applying the same transformation to 123412: - The first digit 1 moves to the second position. - The second digit 2 moves to the third position. - The third digit 3 stays in the same position. - The fourth digit 4 moves to the fifth position. - The fifth digit 1 moves to the sixth position. - The sixth digit 2 moves to the first position. Thus, the transformed number is 312214.

**Q93. Solution**

**Correct Answer: (D)**

Set created out of alphabet of word 'MONO- TONE' is $\{\mathrm{M}, \mathrm{N}, \mathrm{O}, \mathrm{T}, \mathrm{E}\}$ $\therefore$ Total number of subsets $= 2^5 = 32$

**Q94. Solution**

**Correct Answer: (B)**

$\because \quad \tan 15^\circ = \tan(45-30)^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} \therefore \quad \tan 15^\circ + \tan 30^\circ = -p \Rightarrow \quad p = \frac{-4}{\sqrt{3}(\sqrt{3}+1)}$

$q = \tan 15^\circ \times \tan 30^\circ = \frac{\sqrt{3}-1}{\sqrt{3}(\sqrt{3}+1)} \Rightarrow q = \frac{\sqrt{3}-1}{\sqrt{3}(\sqrt{3}+1)} \therefore p \cdot q = \frac{-4}{\sqrt{3}(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{\sqrt{3}(\sqrt{3}+1)}$

$\Rightarrow \quad p \cdot q = \frac{-4(\sqrt{3}-1)}{3(\sqrt{3}+1)^2} = \frac{10-6\sqrt{3}}{3} \Rightarrow p \cdot q = \frac{10-6\sqrt{3}}{3} \Rightarrow$ Option (2) is correct.

**Q95. Solution**

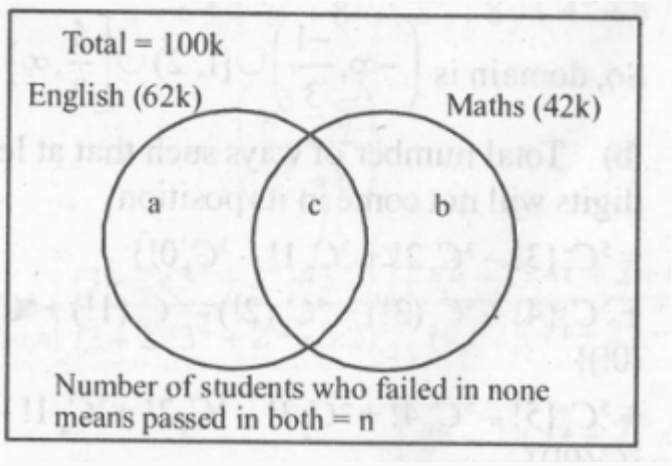
**Correct Answer: (D)**

$|8z_2z_3 + 27z_3z_1 + 64z_1z_2| = |z_1|\,|z_2|\,|z_3| \ \frac{8}{z_1} + \frac{27}{z_2} + \frac{64}{z_3} \ = (2)(3)(4) \ \frac{8\bar{z}_1}{|z_1|^2} + \frac{27\bar{z}_2}{|z_2|^2} + \frac{64\bar{z}_3}{|z_3|^2}$

$= 24\,|2\bar{z}_1 + 3\bar{z}_2 + 4\bar{z}_3| = 24 \ \overline{2z_1 + 3z_2 + 4z_3} \ = 24\,|2z_1 + 3z_2 + 4z_3| = (24)(4) = 96$

**Q96. Solution**

**Correct Answer: (B)**



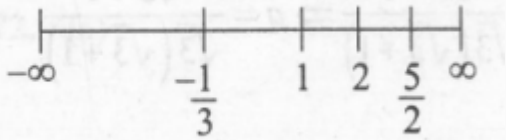

From the given condition, $\mathrm{a}+\mathrm{c}=62\mathrm{k}$; $\mathrm{b}+\mathrm{c}=42\mathrm{k}$ and $\mathrm{c}=20\mathrm{k}$ Hence $\mathrm{a}=42\mathrm{k}$ and $\mathrm{b}=22\mathrm{k}$ Number of students who failed in none means passed in both $=\mathrm{n}=100\mathrm{k}-(\mathrm{a}+\mathrm{b}+\mathrm{c})$
$=100\mathrm{k}-(42\mathrm{k}+22\mathrm{k}+20\mathrm{k})=16\mathrm{k}$. or $16\%$

**Q97. Solution**

**Correct Answer: (A)**

Given function $f(x)=\sqrt{\frac{2x^2-7x+5}{3x^2-5x-2}}$ Here, f(x) should be greater than or equal to 0 . So, $2x^2-7x+5=0$
$\Rightarrow \quad 2x^2-5x-2x+5=0 \Rightarrow \quad (x-1)(2x-5)=0 \Rightarrow \quad x=1, \frac{5}{2}$
$3x^2-5x-2=0 \Rightarrow 3x^2-6x+x-2=0$ $3x(x-2)+1(x-2)=0 \Rightarrow (x-2)(3x+1)=0$

Now,




$\Rightarrow \quad x=2, \frac{-1}{3}$
When we include $x=\frac{-1}{3}, 2$ then $f(x)$ would give not define value so, we will exclude these values from the domain. When we take values between $\left(-\frac{1}{3}, 1\right)$ and So, domain is $\left(-\infty, \frac{-1}{3}\right) \cup[1,2) \cup\left[\frac{5}{2}, \infty\right)$

**Q98. Solution**

**Correct Answer: (B)**

Total number of ways such that at least 3 digits will not come in its position
$={}^5C_3\left\{3!-{}^3C_1 2!+{}^3C_2 1!-{}^3C_3 0!\right\}+{}^5C_4\left\{4!-{}^4C_1(3!)+{}^4C_2(2!)-{}^4C_3(1!)+{}^4C_4(0!)\right\}$
$+{}^5C_4\left\{5!-{}^5C_1 4!+{}^5C_7 3!-{}^5C_2 2!+{}^5C_4 1!-{}^5C_5(0!)\right\}=10(2)+5(9)+(44)=20+45+44=109$

**Q99. Solution**

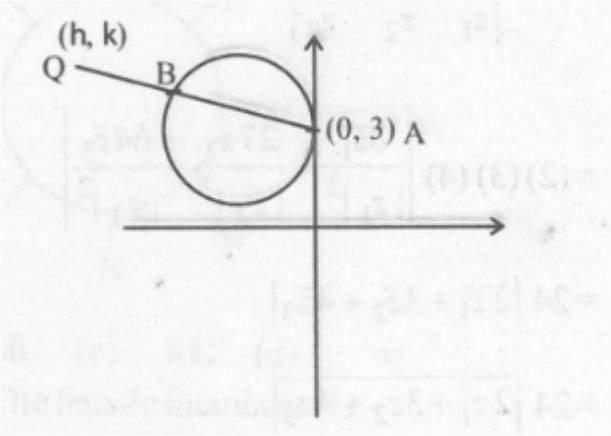
**Correct Answer: (D)**

Given, $17^{\text{th}}$ and $18^{\text{th}}$ terms in the expansion $(2+a)^{50}$ are equal $\therefore T_{17}=T_{18}$
$\Rightarrow {}^{50}C_{16}(2)^{34}(a)^{16}={}^{50}C_{17}(2)^{33}(a)^{17} \Rightarrow a=\frac{{}^{50}C_{16}}{{}^{50}C_{17}} \times 2=1 \therefore$ Now, coefficient of $x^{35}$ in the expansion $(1+x)^{-2}=-36 \because$ Coefficient of $x^r$ is $(r+1)$ in $(1-x)^{-2}$.

**Q100. Solution**

**Correct Answer: (A)**

Given equation of circle $(x+2)^2+(y-3)^2=4$ Let the coordinates of Q is $(h,k)$.




Coordinate of B which is midpoint of AQ because $\mathrm{AQ}=2\mathrm{AB}$. Then, $\mathrm{B}=\left(\frac{0+h}{2},\frac{k+3}{2}\right)\rightarrow\left(\frac{h}{2},\frac{k+3}{2}\right)$ Point B also satisfy the equation of circle. $(x+2)^2+(y-3)^2=4$ $\left(\frac{h}{2}+2\right)^2+\left(\frac{k+3}{2}-3\right)^2=4$ $\frac{(h+4)^2}{4}+\frac{(k-3)^2}{4}=4$ $(h+4)^2+(k-3)^2=16$ Replace $(h,k)$ by $(x,y)$, then, the required equation is $(x+4)^2+(y-3)^2=16$

**Q101. Solution**

**Correct Answer: (B)**

$-10\leq 8\sin\theta+6\cos\theta\leq 10\Rightarrow 0\leq(8\sin\theta+6\cos\theta)^2\leq 100\Rightarrow 2\leq(8\sin\theta+6\cos\theta)^2+2\leq 102$

**Q102. Solution**

**Correct Answer: (C)**

As we know that the image of $(1,1)$ with respect to line $x+y+5=0$ is $\frac{x-1}{1}=\frac{y-1}{1}=\frac{2(1+1+5)}{1+1}$ $\Rightarrow x-1=-7, y-1=-7\Rightarrow x=-6, y=-6\therefore$ Image of point $(1,1)$ is $(-6,-6)$ Now, distance from origin This is the required transformed equation. $D=\sqrt{(0+6)^2+(0+6)^2}$ $D=\sqrt{72}=6\sqrt{2}$

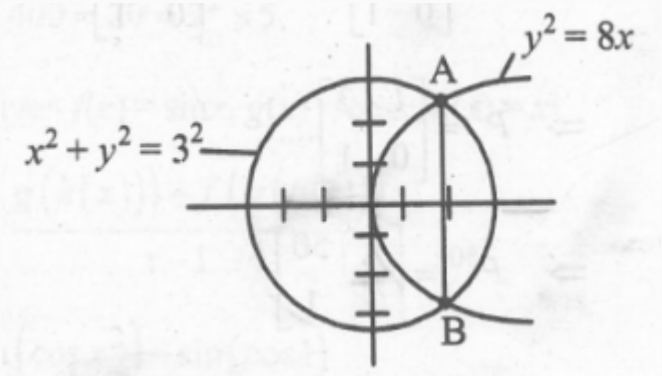
**Q103. Solution**

**Correct Answer: (A)**

By the given condition, $\frac{a+b}{2}=2\sqrt{ab}\Rightarrow a+b=4\sqrt{ab}$ Now, $(a-b)^2=(a+b)^2-4ab=16ab-4ab=12ab\therefore a-b=\sqrt{12ab}=2\sqrt{3}\sqrt{ab}$. (Taking + ve sign only as $a>b$) $\therefore\frac{a+b}{a-b}=\frac{4\sqrt{ab}}{2\sqrt{3}\sqrt{ab}}=\frac{2}{\sqrt{3}}$ By componendo and dividendo, $\frac{2a}{2b}=\frac{2+\sqrt{3}}{2-\sqrt{3}}$ or $\frac{a}{b}=\frac{2+\sqrt{3}}{2-\sqrt{3}}$

**Q104. Solution**

**Correct Answer: (C)**

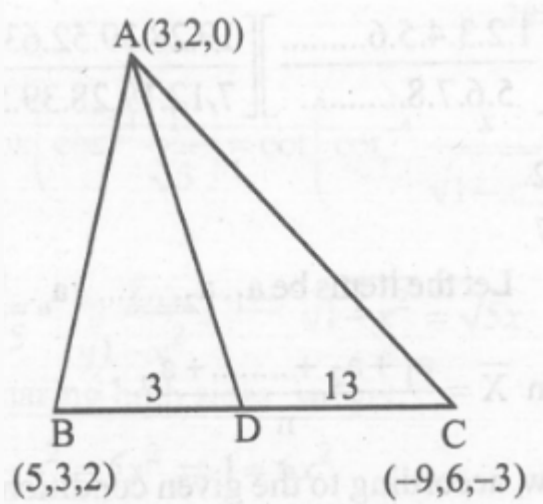



We have : $x^2 + (8x) = 9 \Rightarrow x^2 + 9x - x - 9 = 0 \Rightarrow x(x+9) - 1(x+9) = 0$
$\Rightarrow (x+9)(x-1) = 0 \quad \Rightarrow x = -9, 1$ for $x = 1, y = \pm 2\sqrt{2x} = \pm 2\sqrt{2}$
$\sqrt{(2\sqrt{2} + 2\sqrt{2})^2 + (1-1)^2} = 4\sqrt{2}\ \mathrm{L}_n =$ Length of latus rectum $= 4a = 4 \times 2 = 8\ \mathrm{L}_1 < \mathrm{L}_2$

**Q105. Solution**

**Correct Answer: (B)**

Since $AD$ is the bisector of $\angle BAC$. $\Rightarrow \frac{\mathrm{BD}}{\mathrm{DC}} = \frac{\mathrm{AB}}{\mathrm{AC}}$ ....(i) Now, $\mathrm{AB} = \sqrt{(5-3)^2 + (3-2)^2 + (2-0)^2}$




$= \sqrt{4+1+4} = \sqrt{9} = 3$
$AC = \sqrt{(-9-3)^2 + (6-2)^2 + (-3-0)^2} = \sqrt{144+16+9} = 13 \therefore \quad \frac{\mathrm{BD}}{\mathrm{DC}} = \frac{3}{13}$ Hence, D divides BC in the ratio $3:13$ The co-ordinates of D are $\left(\frac{3(-9)+13(5)}{3+13}, \frac{3(6)+13(3)}{3+13}, \frac{3(-3)+13(2)}{3+13}\right) = \left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$

**Q106. Solution**

**Correct Answer: (A)**

$$\lim_{n\to\infty}\prod_{r=3}^{\infty}\frac{r^3-8}{r^3+8}=\left(\frac{3^3-8}{3^3+8}\right)\left(\frac{4^3-8}{4^3+8}\right)=\lim_{n\to\infty}\left[\frac{(3-2)(3^2+2^2+3.2)}{(3+2)(3^2+2^2-3.2)}\right]\left[\frac{(4-2)(4^2+2+4.2)}{(4+2)(4^2+2^2-4.2)}\right]$$

$$\ldots\ldots\ldots\left(\frac{n^3-8}{n^3+8}\right)\qquad\ldots\ldots\ldots\ldots\left[\frac{(n-2)(n^2+2^n+n.2)}{(n+2)(n^2+2^2-n.2)}\right]$$

$$=\lim_{n\to\infty}\left[\frac{(3-2)(4-2)\ldots\ldots\ldots(n-2)}{(3+2)(4+2)\ldots\ldots\ldots(n+2)}\right]$$

$$\left[\frac{(3^2+2^2+3.2)\,(4^2+2^2+4.2)\ldots\ldots\ldots(n^2+2^2+n.2)}{(3^2+2^2-3.2)\,(4^2+2^2-4.2)\ldots\ldots\ldots(n^2+2^2-n.2)}\right]=\left[\frac{1\cdot2\cdot3.4.5.6.\ldots\ldots\ldots}{5\cdot6.7.8.\ldots\ldots\ldots}\right]\left[\frac{19\cdot28\cdot39.52.63.\ldots\ldots\ldots}{7\cdot12\cdot19.28.39.52.\ldots\ldots}\right]$$

$=\frac{2}{7}$

**Q107. Solution**

**Correct Answer: (C)**

Let the items be $\mathrm{a}_1,\mathrm{a}_2,\ldots\ldots\ldots,\mathrm{a}_\mathrm{n}$ then $\bar{X}=\frac{a_1+a_2+\ldots\ldots\ldots+a_n}{n}$ Now, according to the given condition: $\bar{X}=\frac{(a_1+1)+(a_2+2)+\ldots\ldots\ldots+(a_n+n)}{n}=\overline{\mathrm{X}}+\frac{1+2+3+\ldots\ldots\ldots+n}{n}=\bar{X}+\frac{n(n+1)}{2n}$ (using sum of n natural nos.) $=\overline{\mathrm{X}}+\frac{\mathrm{n}+1}{2}$.

**Q108. Solution**

**Correct Answer: (B)**

$n(S)=$ Total number of ways $=1000$ The favourable cases that the sum of the digits of the marked number on the page is equal to 9 are one digit number or two digits numbers or three digits numbers, if three digit number is $abc$. Then, $a+b+c=9; 0\le a,b,c\le 9.\therefore n(E)=$ Number of favourable ways $=$ Number of solutions of the equation $={}^{9+3-1}C_{3-1}={}^{11}C_2=55\therefore$ Required probability $=\frac{n(E)}{n(S)}=\frac{55}{1000}=\frac{11}{200}$.

**Q109. Solution**

**Correct Answer: (C)**

$f(x)=\frac{x+3}{x-2}\because\quad f(x)$ is not defined for $x=2$ i.e. domain of $f(x)$ is $\mathrm{R}-\{2\}\therefore l=2$ Now, $y=\frac{x+3}{x-2}$ $xy-2y=x+3\ x(y-1)=2y+3\ x=\frac{2y+3}{y-1}$ $y$ can take any value except $y=1$ Co-domain $=\mathrm{R}-\{1\}$ $m=1\ 3l+2m=3(2)+2(1)=8$

**Q110. Solution**

**Correct Answer: (D)**

Since $AA^T=I$ therefore matrix $A$ is orthogonal matrix. Now, $A^TX^{50}A=A^TX^{49}\left(APA^T\right)A$ $=A^TX^{49}AP\left(A^TA\right)=A^TX^{49}AP=A^TX^{48}\left(APA^T\right)AP=A^TX^{48}AP^2\ldots\ldots$ $=A^TAP^{50}=IP^{50}=P^{50}\because\quad P=\begin{bmatrix}1&1\\0&1\end{bmatrix}\Rightarrow P^2=\begin{bmatrix}1&2\\0&1\end{bmatrix}\Rightarrow P^3=\begin{bmatrix}1&3\\0&1\end{bmatrix}\ldots\ldots\Rightarrow\quad P^{50}=\begin{bmatrix}1&50\\0&1\end{bmatrix}$

So, $A^TX^{50}A=P^{50}=\begin{bmatrix}1&50\\0&1\end{bmatrix}$

**Q111. Solution**

**Correct Answer: (B)**

Let the observations be $x_1, x_2, \ldots, x_{20}$ and $\bar{x}$ be their mean. Given that, variance $= 5$ and $n = 20$. We know that, Variance $\left(\sigma^2\right) = \frac{1}{n} \sum_{i=1}^{20} \left(x_i - \bar{x}\right)^2$ i.e. $5 = \frac{1}{20} \sum_{i=1}^{20} \left(x_i - \bar{x}\right)^2$ or $\sum_{i=1}^{20} \left(x_i - \bar{x}\right)^2 = 100$ .....(i) If each observation is multiplied by 2 and the new resulting observations are $y_i$, then $y_i = 2x_i$ i.e., $x_i = \frac{1}{2} y_i$ Therefore, $\bar{\mathrm{y}} = \frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{20} \mathrm{y_i} = \frac{1}{20} \sum_{\mathrm{i}=1}^{20} 2\mathrm{x_i} = 2 \cdot \frac{1}{20} \sum_{\mathrm{i}=1}^{20} \mathrm{x_i}$ i.e., $\bar{y} = 2\bar{x}$ or $\bar{x} = \frac{1}{2}\bar{y}$. On substituting the values of $x_i$ and $\bar{x}$ in eq. (i), we get $\sum_{i=1}^{20} \left(\frac{1}{2} y_i - \frac{1}{2}\bar{y}\right)^2 = 100$ i.e., $\sum_{i=1}^{20} \left(y_i - \bar{y}\right)^2 = 400$ Thus, the variance of new observations $= \frac{1}{20} \times 400 = 20 = 2^2 \times 5$.

**Q112. Solution**

**Correct Answer: (A)**

We have $\cos\left(\cot^{-1}\left(\frac{1}{2}\right)\right) = \cot\left(\cos^{-1} x\right)$ Let $\cot^{-1}\left(\frac{1}{2}\right) = \alpha \Rightarrow \cot\alpha = \frac{1}{2} \Rightarrow \cos^{-1}\frac{1}{\sqrt{5}}$
$\Rightarrow \cos\left(\cos^{-1}\frac{1}{\sqrt{5}}\right) = \cot\left(\cot^{-1}\frac{x}{\sqrt{1-x^2}}\right) \Rightarrow \frac{1}{\sqrt{5}} = \frac{x}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2} = \sqrt{5}x$ On squaring both sides, we get, $1 - x^2 = 5x^2 \Rightarrow 1 = 6x^2 \Rightarrow x = \pm\frac{1}{\sqrt{6}} \quad x = \frac{1}{\sqrt{6}} \quad$ [neglecting - ve sign]

**Q113. Solution**

**Correct Answer: (B)**

Given $f(x) = \sin x, g(x) = \cos x, h(x) = x^2$ $\lim_{x\to 1} \frac{f(g(h(x))) - f(g(h(1)))}{x-1}$ When limit it applied, it gives $\frac{0}{0}$ form, so we apply L'Hospital rule. $\Rightarrow \lim_{x\to 1} \frac{\cos(\cos x^2)(-\sin x^2)(2x) - 0}{1-0}$ Apply the limit,
$\Rightarrow -2(1)\sin(1)\cos(\cos 1) = -2\sin(1)\cos(\cos 1)$

**Q114. Solution**

**Correct Answer: (B)**

For getting infinite solutions $\mathrm{D} = 0, \mathrm{D}_1 = \mathrm{D}_2 = \mathrm{D} = 0$ then check all the three equations Let

$$\Delta = \begin{matrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 1 & 4 & \delta \end{matrix} = 0 \quad \Rightarrow \delta = -3 \text{ And } \Delta_1 = \begin{matrix} 7 & 1 & -1 \\ 1 & -3 & 2 \\ k & 4 & -3 \end{matrix} = 0 \Rightarrow k = 6 \Rightarrow \delta + k = 3$$

**Q115. Solution**

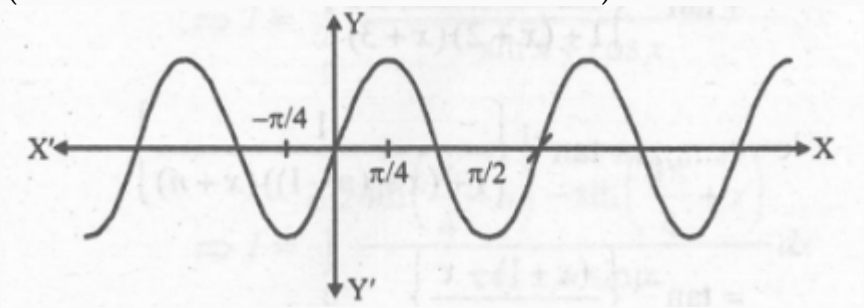
**Correct Answer: (C)**

$$\begin{aligned} \operatorname{adj}\left(\operatorname{adj} A^2\right) &= |\operatorname{adj} \mathrm{A}| = |\mathrm{A}|^{n-1} = |\mathrm{A}|^2 \quad \operatorname{adj} A^2 = |\operatorname{adj} A|^2 = \left(|A|^2\right)^2 = |A|^4 \\ \operatorname{adj}\left(\operatorname{adj} A^2\right) &= \left(|A|^4\right)^{3-1} = \left(|A|^4\right)^2 = |A|^8 \end{aligned}$$

**Q116. Solution**

**Correct Answer: (A)**

Given that, $f(x)=\sin x+\cos x\ g(x)=x^2-1\ g[f(x)]=(\sin x+\cos x)^2-1$
$=\sin^2 x+\cos^2 x+2\sin x\cos x-1=1+\sin 2x-1=\sin 2x$
$(\because \sin^2 x+\cos^2 x=1, \sin x=2\sin x\cos x)$




Among the given options, $\sin 2x$ is monotonous (here strictly increasing) in $-\frac{\pi}{4}\leq x\leq\frac{\pi}{4}$. So, $g(f(x))$ is invertible in $-\frac{\pi}{4}\leq x\leq\frac{\pi}{4}$.

**Q117. Solution**

**Correct Answer: (A)**

Here, $f(x)$ is continuous on the interval $[0,8]$. So it will also be continuous on 2 and 4. At $x=2$, $\lim_{x\to 2^-} f(x)=\lim_{x\to 2^+} f(x)\ \lim_{x\to 2^-}\left(x^2+\mathrm{ax}+\mathrm{b}\right)=\lim_{x\to 2^+}(3x+2)\ 4+2a+b=3(2)+2$ $\therefore 2a+b=4$ ...(i) At $x=4\ \lim_{x\to 4^-} f(x)=\lim_{x\to 4^+} f(x)\ \lim_{x\to 4^-}(3x+2)=\lim_{x\to 4^+} 2ax+5b$ $3(4)+2=2a(4)+5b\ \therefore 8a+5b=14$ .....(ii) On solving Eqs. (i) and (ii), we get $a=3$ and $b=-2$.

**Q118. Solution**

**Correct Answer: (B)**

Given, $y=\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+\cdots$ to $n$ terms
$=\tan^{-1}\left\{\frac{1}{1+x(x+1)}\right\}+\tan^{-1}\left\{\frac{1}{1+(x+1)(x+2)}\right\}+\tan^{-1}\left\{\frac{1}{1+(x+2)(x+3)}\right\}$
$+\ldots\ldots\ldots+\tan^{-1}\left\{\frac{1}{1+(x+(n-1))(x+n)}\right\}=\tan^{-1}\left\{\frac{(x+1)-x}{1+(x+1)x}\right\}+\tan^{-1}\left\{\frac{(x+2)-(x+1)}{1+(x+2)(x+1)}\right\}$
$+\tan^{-1}\left\{\frac{(x+3)-(x+2)}{1+(x+3)(x+2)}\right\}+\ldots.+\tan^{-1}\left(\frac{(x+n)-(x+n-1)}{1+(x+n)(x+n-1)}\right)\therefore y=\left\{\tan^{-1}(x+1)-\tan^{-1}(x)\right\}$
$+\left\{\tan^{-1}(x+2)-\tan^{-1}(x+1)\right\}+\left\{\tan^{-1}(x+3)-\tan^{-1}(x+2)\right\}$
$+\ldots\ldots\ldots+\left\{\tan^{-1}(x+n)-\tan^{-1}[x+(n-1)]\right\}$ So, $\quad y=\tan^{-1}(x+n)-\tan^{-1}(x)$ On differentiating both sides w.r.t. $x$, we get $\frac{dy}{dx}=\frac{1}{1+(x+n)^2}-\frac{1}{1+x^2}$

**Q119. Solution**

**Correct Answer: (A)**

Correct result is $(\sim \mathrm{p}\vee\sim \mathrm{q})\Rightarrow(\mathrm{r}\wedge \mathrm{s})$ So, $\sim(p\wedge q)\Rightarrow(r\wedge s)$

**Q120. Solution**

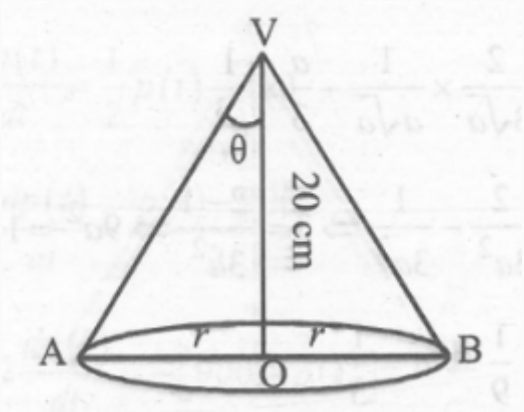
**Correct Answer: (C)**

Given that $g(u)=2\tan^{-1}(e^u)-\frac{\pi}{2}\therefore\quad g(-u)=2\tan^{-1}(e^{-u})-\frac{\pi}{2}=2\tan^{-1}\left(\frac{1}{e^u}\right)-\frac{\pi}{2}$
$=2\cot^{-1}(e^u)-\frac{\pi}{2}=2\left[\frac{\pi}{2}-\tan^{-1}(e^u)\right]-\frac{\pi}{2}=\pi-2\tan^{-1}(e^u)-\frac{\pi}{2}=\frac{\pi}{2}-2\tan^{-1}(e^u)=-g(u)$
$\therefore\quad g$ is an odd function. Also $g'(u)=\frac{2e^u}{1+e^{2u}}>0,\forall u\in(-\infty,\infty)\therefore g$ is strictly increasing on $(-\infty,\infty)$.

**Q121. Solution**

**Correct Answer: (B)**

Let $\theta$ be the semi-vertical angle and $r$ be the radius of the cone at time $t$. the radius of the cone at time $t$. Then,




$r=20\tan\theta$
$\Rightarrow\quad \frac{dr}{dt}=20\sec^2\theta\frac{d\theta}{dt}\Rightarrow\quad \frac{dr}{dt}=20\sec^2 30^\circ\times 2\left[\because\theta=30^\circ\text{ and }\frac{d\theta}{dt}=2\right]$
$\Rightarrow\quad \frac{dr}{dt}=20\times\frac{4}{3}\times 2\text{ cm/s}=\frac{160}{3}\text{ cm/s}$

**Q122. Solution**

**Correct Answer: (D)**

$\int\sqrt{x+\sqrt{x^2+2}}dx\ \sqrt{x+\sqrt{x^2+2}}=t\Rightarrow x+\sqrt{x^2+2}=t^2\ \sqrt{x^2+2}=t^2-x$
$\Rightarrow x^2+2=t^4+x^2-2t^2x\Rightarrow \mathrm{x}=\frac{\mathrm{t}^4-2}{2\mathrm{t}^2}\Rightarrow \mathrm{dx}=\frac{\mathrm{t}^4+2}{\mathrm{t}^3}\mathrm{dt}\int\mathrm{t}\cdot\frac{\mathrm{t}^4+2}{\mathrm{t}^3}\mathrm{dt}=\int\left(\mathrm{t}^2+\frac{2}{\mathrm{t}^2}\right)\mathrm{dt}=\frac{\mathrm{t}^3}{3}-\frac{2}{\mathrm{t}}+\mathrm{C}$
$=\frac{t^3}{3}-\frac{2}{t}+C=\frac{t^4-6}{3t}+C=\frac{\left(x+\sqrt{x^2+2}\right)^2-6}{3\sqrt{x+\sqrt{x^2+2}}}+C$

**Q123. Solution**

**Correct Answer: (C)**

Here $f(x)$ is continous $\forall x\in\mathbf{R}$ Now At $x=-\sqrt{5}$ L.H.D = 0 R.H.D $=2x=-2\sqrt{5}\Rightarrow\mathrm{f(x)}$ is not differentiable at $\mathrm{x}=-\sqrt{5}$ At $x=\sqrt{5}$ L.H.D. $=2x=2\sqrt{5}$ R.H.D. $=0\Rightarrow\mathrm{f(x)}$ is not differentiable at $\mathrm{x}=\sqrt{5}$ $\Rightarrow\mathrm{k}=2\ k-2=2-2=0$

**Q124. Solution**

**Correct Answer: (A)**

Here $\frac{d^2y}{dx^2}=n(n-1)(x-a)^{n-2}$ Now, $\frac{d^2y}{dx^2}=0\Rightarrow x=a$ Differentiating equation of the curve $n$ times, we get, $\frac{d^ny}{dx^n}=n\,!\therefore\quad$ at $x=a,\frac{d^ny}{dx^n}\neq 0$ and $\frac{d^{n-1}y}{dx^{n-1}}=0$, where $n$ is odd. Therefore $(a,0)$ is the point of inflexion

**Q125. Solution**

**Correct Answer: (B)**

Let $I=\int_0^{\frac{\pi}{2}} \frac{\sin\left(\frac{\pi}{4}+x\right)+\sin\left(\frac{3\pi}{4}+x\right)}{\cos x+\sin x} dx \Rightarrow I=\int_0^{\pi/2} \frac{\sin\left(\frac{3\pi}{4}-x\right)+\sin\left(\frac{5\pi}{4}-x\right)}{\sin x+\cos x} dx \Rightarrow I=\int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{4}+x\right)-\sin\left(\frac{3\pi}{4}+x\right)}{\cos x+\sin x} dx$ Now $I+I=\int_0^{\pi/2} \frac{2\sin\left(\frac{\pi}{4}+x\right)}{\cos x+\sin x} dx \Rightarrow I=\int_0^{\pi/2} \frac{\frac{1}{\sqrt{2}}\cos x+\frac{1}{\sqrt{2}}\sin x}{\cos x-\sin x} dx=\frac{\pi}{2\sqrt{2}}$

**Q126. Solution**

**Correct Answer: (C)**

Given statement $S_1:(\sim p \vee q) \vee (\sim p \vee r) \equiv \sim p \vee (q \vee r)$ $S_2: p \rightarrow (q \vee r) \equiv \sim p \vee (q \vee r) \rightarrow$ By conditional law $S_1 \equiv S_2$ Option (1): If S2 is true, then S1 is true. - If S2 is true, then $\sim$ p $\vee$ (q $\vee$ $r$) is true. This implies that either $p$ is false, or $q \vee r$ is true. In both cases, S 1 will also be true because $\sim$ p $\vee$ q $\vee$ r is true if either $\sim$ p or q $\vee$ r is true. - This is true. Option (2): If S2 is false, then S1 is false. - If S 2 is false, then $p$ must be true and $q \vee r$ must be false. In that case, S 1 will be $\sim$ p $\vee$ $q \vee r$, which simplifies to false because $p$ is true and both $q$ and $r$ are false. - This is true. Option (3): If S2 is false, then S1 is true. - If S2 is false, then $p$ is true and q $\vee$ $r$ is false. So, S1 will be $\sim$ p $\vee$ q $\vee$ $r$, which simplifies to false because $p$ is true and both $q$ and $r$ are false. Thus, S 1 is not true. - This is NOT true. Option (4): If S1 is false, then S2 is false. - If S1 is false, then $\sim p \vee q \vee r$ is false, which implies that $p$ is true, and both $q$ and $r$ are false. In that case, $\mathbf{S2}(\sim p \vee (q \vee r))$ will be false because $p$ is true and both $q$ and $r$ are false. - This is true. Answer: The correct answer is (3) If S2 is false, then S1 is true. This statement is NOT true.

**Q127. Solution**

**Correct Answer: (D)**

Let $I=\int e^{\tan\theta}(\sec\theta-\sin\theta)d\theta$ Put $\tan\theta=t \Rightarrow \sec^2\theta d\theta=dt \Rightarrow d\theta=\frac{dt}{1+t^2}$
$\Rightarrow I=\int e^t\left(\sqrt{1+t^2}-\frac{t}{\sqrt{1+t^2}}\right)\frac{dt}{1+t^2}=\int e^t\left(\frac{1}{\sqrt{1+t^2}}-\frac{t}{(1+t^2)^{3/2}}\right)dt$ Integrating first part by parts we have,
$=\frac{1}{\sqrt{1+t^2}}e^t+\int\frac{t}{(1+t^2)^{3/2}}\cdot e^t dt-\int\frac{t}{(1+t^2)^{3/2}}e^t dt+c=\frac{e^t}{\sqrt{1+t^2}}+c=e^{\tan\theta}\cos\theta+c$

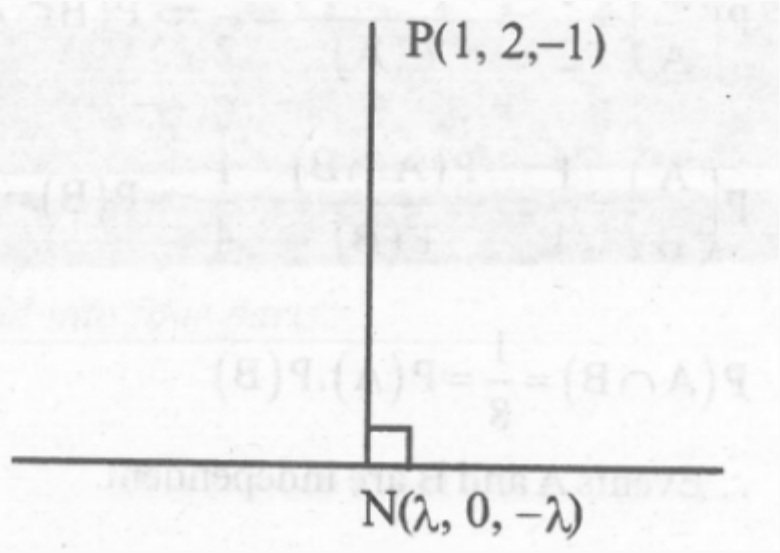
**Q128. Solution**

**Correct Answer: (A)**

Given differential equation is $\frac{dp(t)}{dt}=0.5p(t)-450 \Rightarrow \frac{dp(t)}{dt}=\frac{1}{2}p(t)-450 \Rightarrow \quad \frac{dp(t)}{dt}=\frac{p(t)-900}{2}$
$\Rightarrow \quad 2\frac{dp(t)}{dt}=-[900-p(t)] \Rightarrow \quad 2\frac{dp(t)}{900-p(t)}=-dt$ Integrate both the side, we get : $-2\int\frac{dp(t)}{900-p(t)}=\int dt$ Let $900-p(t)=u \Rightarrow -dp(t)=du \therefore$ We have; $2\int\frac{du}{u}=\int dt \Rightarrow 2\ln u=t+c$
$\Rightarrow \quad 2\ln[900-p(t)]=t+c$ when $t=0, p(0)=850$ $2\ln(50)=c \quad \Rightarrow 2\left[\ln\left(\frac{900-p(t)}{50}\right)\right]=t$
$\Rightarrow \quad 900-p(t)=50e^{\frac{t}{2}} \Rightarrow \quad p(t)=900-50e^{\frac{t}{2}}$ let $p(t_1)=0$ $0=900-50e^{\frac{t_1}{2}} \quad \therefore t_1=2\ln 18$

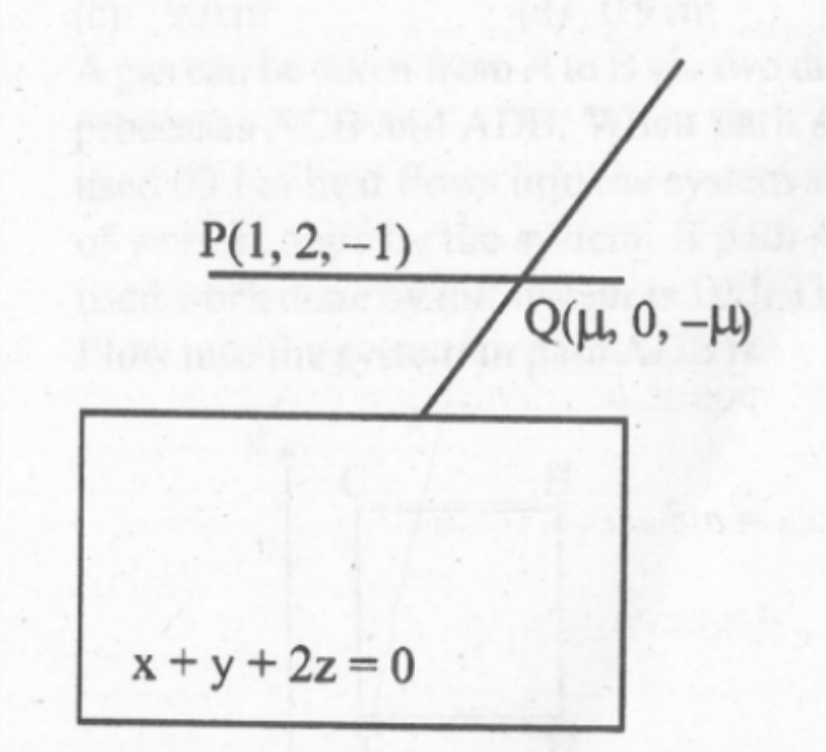
**Q129. Solution**

**Correct Answer: (C)**




Let $\frac{x}{1}=\frac{y}{0}=\frac{z}{-1}=\lambda \Rightarrow \mathrm{N}(\lambda, 0,-\lambda)\ \overrightarrow{\mathrm{b}}=\hat{\mathrm{i}}-\hat{\mathrm{k}}$

$\because \overrightarrow{\mathrm{PN}} \cdot \overrightarrow{\mathrm{b}}=0 \Rightarrow 1(1-\lambda)-(\lambda-1)=0 \Rightarrow \lambda=1 \Rightarrow \mathrm{N}(1,0,-1)$ Let $\mathrm{Q}(\mu, 0,-\mu) \because \overrightarrow{\mathrm{n}}=\hat{\mathrm{i}}+\hat{\mathrm{j}}+2\hat{\mathrm{k}}$ Now,




$\because \overrightarrow{\mathrm{PQ}} \cdot \overrightarrow{\mathrm{n}}=0 \Rightarrow \mu=-1 \Rightarrow \mathrm{Q}(-1,0,1)\ \overrightarrow{\mathrm{PN}}=2\hat{\mathrm{j}}$ and $\overrightarrow{\mathrm{PQ}}=2\hat{i}+2\hat{j}-2\hat{k} \Rightarrow \cos \alpha=\frac{1}{\sqrt{3}}$

**Q130. Solution**

**Correct Answer: (D)**

$\mathrm{P}\left(\frac{\mathrm{B}}{\mathrm{A}}\right)=\frac{1}{2} \Rightarrow \frac{\mathrm{P}(\mathrm{B} \cap \mathrm{A})}{\mathrm{P}(\mathrm{A})}=\frac{1}{2} \Rightarrow \mathrm{P}(\mathrm{B} \cap \mathrm{A})=\frac{1}{8}\ \mathrm{P}\left(\frac{\mathrm{A}}{\mathrm{B}}\right)=\frac{1}{4} \Rightarrow \frac{\mathrm{P}(\mathrm{A} \cap \mathrm{B})}{\mathrm{P}(\mathrm{B})}=\frac{1}{4} \Rightarrow \mathrm{P}(\mathrm{B})=\frac{1}{2}$

$\mathrm{P}(\mathrm{A} \cap \mathrm{B})=\frac{1}{8}=\mathrm{P}(\mathrm{A}) \cdot \mathrm{P}(\mathrm{B}) \therefore$ Events A and B are independent. Now, $\mathrm{P}\left(\frac{\mathrm{A}'}{\mathrm{B}}\right)=\frac{\mathrm{P}(\mathrm{A} \cap \mathrm{B})}{\mathrm{P}(\mathrm{B})}=\frac{\mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B})}{\mathrm{P}(\mathrm{B})}=\frac{3}{4}$

and $\mathrm{P}\left(\frac{\mathrm{B}'}{\mathrm{A}'}\right)=\frac{\mathrm{P}(\mathrm{B}' \cap \mathrm{A}')}{\mathrm{P}(\mathrm{A}')}=\frac{\mathrm{P}(\mathrm{B}')\mathrm{P}(\mathrm{A}')}{\mathrm{P}(\mathrm{A}')}=\frac{1}{2}$

**Q131. Solution**

**Correct Answer: (A)**

$l = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$

$A = 1 \text{ mm}^2 = 1 \times 10^{-4} \text{ m}^2 \therefore$ Resistance of wire, $R = \frac{V}{i} = \frac{2}{4} = 0.5\Omega$

$i = 4 \text{ A}, V = 2\text{volt}$

$$\therefore \text{ Resistiviry, } \rho = R\frac{A}{l} = 0.5 \times \frac{1 \times 10^{-6}}{50 \times 10^{-2}}$$
$$= \frac{0.5 \times 1 \times 10^{-4}}{50}$$
$$= 1 \times 10^{-6}\Omega - \text{fm}$$

**Q132. Solution**

**Correct Answer: (C)**

Given propositions $\mathrm{P}_1 :\sim (\mathrm{p} \rightarrow \sim \mathrm{q})$ $\mathrm{P}_2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$ Required table is shown below.

| **p** | **q** | $\sim p$ | $\sim q$ | $\sim p \vee q$ | $p \rightarrow (\sim (p \vee q)$ | $p \rightarrow \sim q$ | $\sim (p \rightarrow \sim q)$ | $p \wedge \sim q$ | $p_2$ |
|---|---|---|---|---|---|---|---|---|---|
| T | T | F | F | T | T | F | T | F | F |
| T | F | F | T | F | F | T | F | T | F |
| F | T | T | F | T | T | T | F | F | F |
| F | F | T | T | T | T | T | F | F | F |

$p \rightarrow (\sim p \vee q)$ is $F$ when $p$ is true $q$ is false From table $P_1$ & $P_2$ both are false

**Q133. Solution**

**Correct Answer: (D)**

Alkyl halides react with sodium alkoxide to give ether. This is called Williamson's synthesis of ether.

$$\mathrm{C_2H_5Cl + C_2H_5ONa \xrightarrow{\Delta} C_2H_5OC_2H_5 + NaCl}$$

**Q134. Solution**

**Correct Answer: (B)**

With progressive increase in atomic number, the reduction potential of halogens decreases, thus oxidising power also decreases. Hence, a halogen with lower atomic number will oxidise the halide ion of higher atomic number and therefore, will liberate them from their salt solution. Hence, the reaction

$$\mathrm{Cl_2 + 2\,F^- \longrightarrow 2Cl^- + F_2}$$

is not possible.

**Q135. Solution**

**Correct Answer: (D)**

For equimolar solution, $\chi_b = \chi_t = 0.5$

$p_b = \chi_b \times p_b^0 = 0.5 \times 160 = 80$ mm

$p_{total} = 80 + 30 = 110$ mm Mole fraction of benzene in vapour phase $= \frac{p_b}{p_{total}} = \frac{80}{110} = 0.727 = 0.73$

$p_b = \chi_t \times p_b^0 = 0.5 \times 60 = 30$ mm

**Q136. Solution**

**Correct Answer: (C)**

Ratio of amplitudes, $\frac{a_1}{a_2} = \frac{3}{2} \therefore$

$$a_1 = 3k, a_2 = 2k$$

$$\frac{I_{\max}}{I_{\text{mix}}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(3k + 2k)^2}{(3k - 2k)^2} = \frac{25}{1}$$

$$I_{\text{man}} : I_{\min} = 25 : 1$$

**Q137. Solution**

**Correct Answer: (B)**

$$m = 2.05 \times 10^6 \text{ kg}, v_1 = 5 \text{ m/s}$$

$$v_2 = 25 \text{ m/s}, t = 5 \text{ min} = 5 \times 60 = 300\text{sec}$$

$$\text{Power} = \frac{W}{t} = \frac{\text{change in KE}}{t} = \frac{1}{2} \frac{m\left(v_2^2 - v_1^2\right)}{300} = \frac{1}{2} \times \frac{2.05 \times 10^6 \left[(25)^2 - (5)^2\right]}{300} = \frac{1}{2} \times \frac{2.05 \times 10^6 \times 600}{300} = 2.05 \times 10^6 \text{ W} = 2.05\text{MW}$$

**Q138. Solution**

**Correct Answer: (A)**

To find the alphanumeric relationship in the given question, you have to know the place value of each of the given alphabets. The place value of C is 3, and it is one for A. Similarly, 7 is for G and 5 is for E. As we can see, the given numbers are nothing but the place values of the given alphabets.

31 =CA GE = 75

89 = HI DE= 45

Hence, HIDE is the correct answer to the given question.

**Q139. Solution**

**Correct Answer: (C)**

In the given question the interpretation, since $1975$, there has been in the United States a dramatic decline in the incidence of traditional childhood diseases such as measles that reduced incidence of measles is connected to the increased occurrence of Peterson's disease. Also, this problem is not seen in adults, who probably had measles in their childhood.

Option C talks about the difference in the vulnerable of adults from children because children have poor immunity.

So, option C is the correct answer.

**Q140. Solution**

**Correct Answer: (D)**

We have, $P(X=1)=P(X=2)$

$$\frac{\lambda^1}{1!}e^{\lambda}=\frac{\lambda^2}{2!}e^{\lambda}\Rightarrow\lambda=2$$

Also, $P(X=0)+P(X=1)+P(X=2)=1$

$$\Rightarrow \quad 0.4+P(X=1)+P(X=2)=1$$
$$\Rightarrow \quad P(X=1)+P(X=2)=0.6=\frac{6}{10}=\frac{3}{5}$$

Also, $P(X=1)+P(X=1)=\frac{3}{5}$ [from (i)]

$$\Rightarrow \quad P(X=1)=\frac{3}{10}$$
$$\Rightarrow \quad P(X=1)=P(X=2)=\frac{3}{10}$$

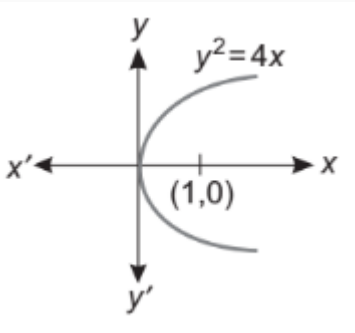
Mean $X_0P(X=0)+X_1P(X=1)+X_2P(X=2)$

$$=0+1\cdot\frac{3}{10}+2\frac{3}{10}=\frac{9}{10}=0.9$$

**Q141. Solution**

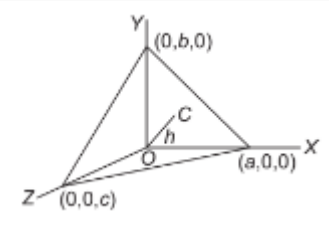
**Correct Answer: (B)**

Given curve is $y^2=4x$. Also, point $(1,0)$ is the focus of the parabola. It is clear from the graph that only one normal is possible.



**Q142. Solution**

**Correct Answer: (D)**




The centroid of the triangle is $C\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$

The direction coines of the line $OC$ are $\left(\frac{h}{a}, \frac{h}{b}, \frac{h}{c}\right)$. As we know that,

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow \quad \frac{h^2}{a^2} + \frac{h^2}{b^2} + \frac{h^2}{c^2} = 1$$

$$\Rightarrow \quad \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{h^2}$$

$$\Rightarrow \quad \left(\frac{3}{a}\right)^2 + \left(\frac{3}{b}\right)^2 + \left(\frac{3}{c}\right)^2 = \frac{9}{h^2}$$

The locus of the centroid is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{h^2}$

C I P H Ξ R the goat 🫡